ACHARYA NARENDRA DEV COLLEGE
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DATA MINING PROJECT

Breast Cancer Diagnosis using Data Mining Techniques

(Wisconsin Diagnostic Breast Cancer Dataset)

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Date:

1. Dataset

- I used the **Breast Cancer Wisconsin (Diagnostic)** dataset. It is a well known dataset from the UCI Machine Learning Repository, and it is also available inside scikit-learn.
- It has **569 patients** and **30 numbers (features)** for each patient. There are no missing values.
- The numbers were measured from the cells of a breast lump, taken with a thin needle (FNA test). Examples are radius, texture, perimeter, area, smoothness and concavity of the cell.
- For each of 10 measurements, the dataset gives 3 values: the mean, the standard error and the worst (largest) value. So 10 x 3 = 30 features.
- The answer we want to predict is the **diagnosis**: malignant (cancer) for 212 patients and benign (not cancer) for 357 patients.

Sample of the dataset (4 malignant and 4 benign patients, 8 of the 32 columns)

Mean radius	Mean texture	Mean perimeter	Mean area	Mean smooth.	Worst radius	Worst concave pts	Diagnosis
17.99	10.38	122.80	1001.0	0.1184	25.38	0.2654	Malignant
20.57	17.77	132.90	1326.0	0.0847	24.99	0.1860	Malignant
19.69	21.25	130.00	1203.0	0.1096	23.57	0.2430	Malignant
11.42	20.38	77.58	386.1	0.1425	14.91	0.2575	Malignant
13.54	14.36	87.46	566.3	0.0978	15.11	0.1288	Benign
13.08	15.71	85.63	520.0	0.1075	14.50	0.0728	Benign
9.50	12.44	60.34	273.9	0.1024	10.23	0.0623	Benign
13.03	18.42	82.61	523.8	0.0898	13.30	0.0501	Benign

The full dataset (569 rows and 32 columns) is submitted with the project as breast_cancer_dataset.csv.

2. Aim and Objectives

Aim: To use data mining to predict whether a tumour is malignant (cancer) or benign (not cancer) from the cell measurements.

Objectives:

- Understand the data and check it for problems.
- Find which measurements are the most useful.
- Train 6 different models and find the best one.
- Group the patients using K-Means clustering and see if the groups match the real diagnosis.

3. Tools

I used Python with these libraries: pandas (to handle the data), NumPy (for maths), scikit-learn (to build the models), and matplotlib and seaborn (to draw the graphs). I wrote and ran the code in VS Code.

4. Theory

- **Data mining** means finding useful patterns in a large amount of data. The steps are: take the data, clean it, apply the techniques, and check the results.

- **Standardisation:** the features have very different sizes (area is in hundreds, smoothness is below 1). So I scaled all of them to the same range (mean 0, standard deviation 1) so that no feature dominates.
- **Train-test split:** 80% of the data is used to train the model and 20% is kept to test it on data it has never seen.
- **Feature selection** means choosing the measurements that help the most in prediction. I used the ANOVA F-test and Random Forest importance.
- **PCA** is a method that reduces many features to a few new features, while keeping most of the information.
- **Classification:** we teach a model using data that already has the answer (label). Then it predicts the answer for new patients. I used Logistic Regression, KNN, Decision Tree, Random Forest, SVM and Naive Bayes.
- **Cross-validation:** the model is tested 5 times on different parts of the data, to check that the result is reliable.
- **K-Means clustering:** it groups similar patients together without using the diagnosis (no labels).

5. Important Definitions

Term	Meaning
Feature	A column of data used for prediction, e.g. mean radius
Target	The thing we want to predict (the diagnosis)
Malignant / Benign	Cancer / not cancer
Overfitting	The model remembers the training data but does badly on new data
Confusion matrix	A table that shows the correct and wrong predictions
Accuracy	Out of all patients, how many were predicted correctly
Precision	Out of the patients predicted positive, how many really are positive
Recall	Out of all the really positive patients, how many we found
F1-score	One score that combines precision and recall
ROC-AUC	How well the model separates the two classes (1 is perfect, 0.5 is just guessing)
Silhouette score	How well a patient fits in its own cluster (from -1 to 1)

6. Code and Output

The full Python code is in the file breast_cancer_data_mining.py, which is submitted with this report. The code runs in 8 steps. The screenshots below show the output I got.

- **Step 1:** Load the dataset (569 rows, 32 columns).
- **Step 2:** Check the data: missing values, number of benign and malignant patients, basic statistics.
- **Step 3:** Preprocessing: split into 80% train and 20% test, then scale the features.
- **Step 4:** Feature selection: find the top 10 features.
- **Step 5:** PCA: reduce the 30 features.
- **Step 6:** Train the 6 models and compare them.
- **Step 7:** K-Means clustering.
- **Step 8:** Final summary.

```
================================================================
1. LOADING THE DATASET
================================================================
Shape (rows, columns): (569, 32)

First 5 rows:
   mean radius  mean texture  mean perimeter  mean area  ...  worst symmetry  worst fractal dimension  target  \
0        17.99         10.38          122.80     1001.0  ...          0.4601                  0.11890       0
1        20.57         17.77          132.90     1326.0  ...          0.2750                  0.08902       0
2        19.69         21.25          130.00     1203.0  ...          0.3613                  0.08758       0
3        11.42         20.38           77.58      386.1  ...          0.6638                  0.17300       0
4        20.29         14.34          135.10     1297.0  ...          0.2364                  0.07678       0

   diagnosis
0  Malignant
1  Malignant
2  Malignant
3  Malignant
4  Malignant

[5 rows x 32 columns]

================================================================
2. EXPLORATORY DATA ANALYSIS
================================================================
Missing values: 0
Duplicate rows: 0

Class distribution:
diagnosis
Benign       357
Malignant    212
Name: count, dtype: int64
```

Output 1: Steps 1-2, loading the dataset and class distribution

```
Statistical summary (first 6 features):
       mean radius  mean texture  mean perimeter  mean area  mean smoothness  mean compactness
count      569.000       569.000         569.000    569.000          569.000           569.000
mean        14.127        19.290          91.969    654.889            0.096             0.104
std          3.524         4.301          24.299    351.914            0.014             0.053
min          6.981         9.710          43.790    143.500            0.053             0.019
25%         11.700        16.170          75.170    420.300            0.086             0.065
50%         13.370        18.840          86.240    551.100            0.096             0.093
75%         15.780        21.800         104.100    782.700            0.105             0.130
max         28.110        39.280         188.500   2501.000            0.163             0.345

Top 5 features most correlated with diagnosis (negative = malignant):
worst concave points   -0.794
worst perimeter        -0.783
mean concave points    -0.777
worst radius           -0.776
mean perimeter         -0.743
dtype: float64

================================================================
3. DATA PREPROCESSING
================================================================
Train size: (455, 30) | Test size: (114, 30)
Features standardised (mean=0, std=1).

================================================================
4. FEATURE SELECTION
================================================================
Top 10 features by ANOVA F-score:
worst concave points    795.22
worst perimeter         719.60
mean concave points     703.36
worst radius            687.43
mean perimeter          565.49
worst area              541.71
mean radius             523.61
```

Output 2: Steps 2-4, statistics, correlation, preprocessing and ANOVA feature scores

```
mean area              469.52
mean concavity         416.37
worst concavity        365.75
dtype: float64

Top 10 features by Random Forest importance:
worst perimeter        0.1373
worst area             0.1373
worst concave points   0.1155
mean concave points    0.0918
worst radius           0.0841
mean radius            0.0627
mean perimeter         0.0497
mean concavity         0.0441
mean area              0.0405
worst concavity        0.0348
dtype: float64

================================================================
5. PRINCIPAL COMPONENT ANALYSIS (PCA)
================================================================
PC1 explains 44.27% variance
PC2 explains 18.97% variance
Components needed for 95% variance: 10 (out of 30)

================================================================
6. CLASSIFICATION MODELS
================================================================
                    Accuracy  Precision  Recall  F1-Score  ROC-AUC  CV Mean Acc  CV Std
Model
Logistic Regression    0.9825     0.9861  0.9861    0.9861   0.9954       0.9780  0.0098
SVM (RBF)              0.9825     0.9861  0.9861    0.9861   0.9950       0.9670  0.0155
KNN (k=5)              0.9561     0.9589  0.9722    0.9655   0.9788       0.9604  0.0112
Random Forest          0.9561     0.9589  0.9722    0.9655   0.9932       0.9626  0.0179
Naive Bayes            0.9298     0.9444  0.9444    0.9444   0.9868       0.9341  0.0287
Decision Tree          0.9211     0.9565  0.9167    0.9362   0.9163       0.9231  0.0120
```

Output 3: Steps 4-6, feature importance, PCA and model comparison

```
BEST MODEL: Logistic Regression

Classification report:
              precision    recall  f1-score   support

   Malignant       0.98      0.98      0.98        42
      Benign       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114

Confusion matrix:
 [[41  1]
 [ 1 71]]

Random Forest with only top-10 features: accuracy = 0.9386

================================================================
7. CLUSTERING - K-MEANS
================================================================
Silhouette score (k=2): 0.3434
Adjusted Rand Index vs true labels: 0.6536
Cluster-vs-diagnosis agreement: 90.51%

Cluster vs actual diagnosis:
Cluster      0    1
Actual
Benign      18  339
Malignant  176   36
```

Output 4: Steps 6-7, best model report, confusion matrix and K-Means

```
================================================================
8. SUMMARY
================================================================
Best classifier        : Logistic Regression
Best test accuracy     : 98.25%
Best ROC-AUC           : 0.9954
PCA components for 95% : 10 of 30
All graphs saved in the 'figures/' folder; table in model_comparison.csv
```

Output 5: Step 8, summary of results

7. Tables and Graphs

Table 1: Results of the six models on the test data

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Acc.
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954	0.9780
SVM (RBF)	0.9825	0.9861	0.9861	0.9861	0.9950	0.9670
KNN (k=5)	0.9561	0.9589	0.9722	0.9655	0.9788	0.9604
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9932	0.9626
Naive Bayes	0.9298	0.9444	0.9444	0.9444	0.9868	0.9341
Decision Tree	0.9211	0.9565	0.9167	0.9362	0.9163	0.9231

Table 2: Confusion matrix of the best model (Logistic Regression)

	Predicted Malignant	Predicted Benign
Actual Malignant	41	1
Actual Benign	1	71

Table 3: K-Means clusters (k = 2) compared with the real diagnosis

Actual diagnosis	Cluster 0	Cluster 1	Total
Benign	18	339	357
Malignant	176	36	212

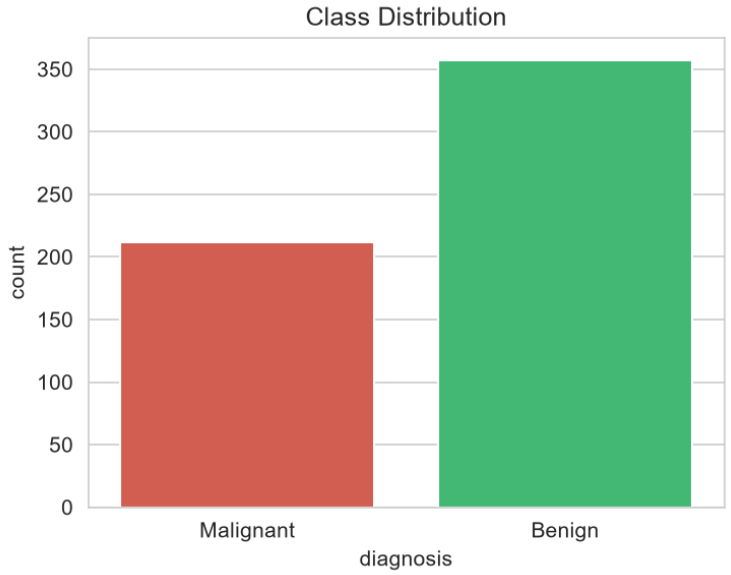


Figure 1: Class distribution (357 benign, 212 malignant)

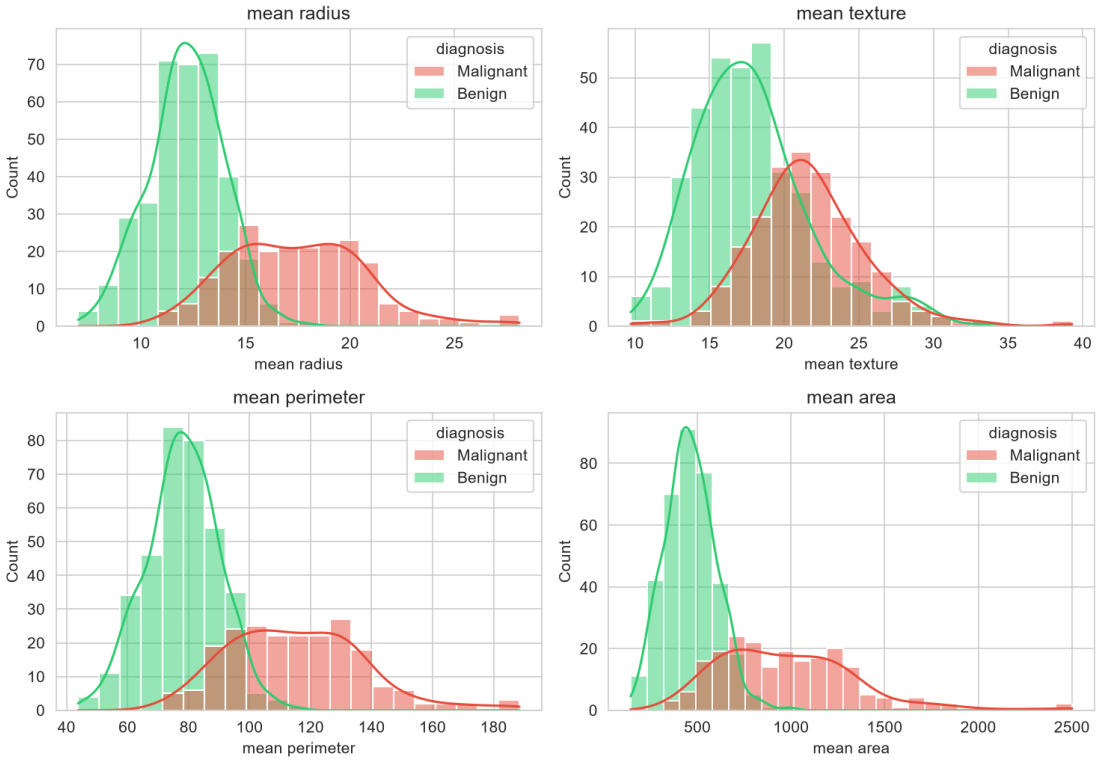


Figure 2: Distribution of key features by diagnosis

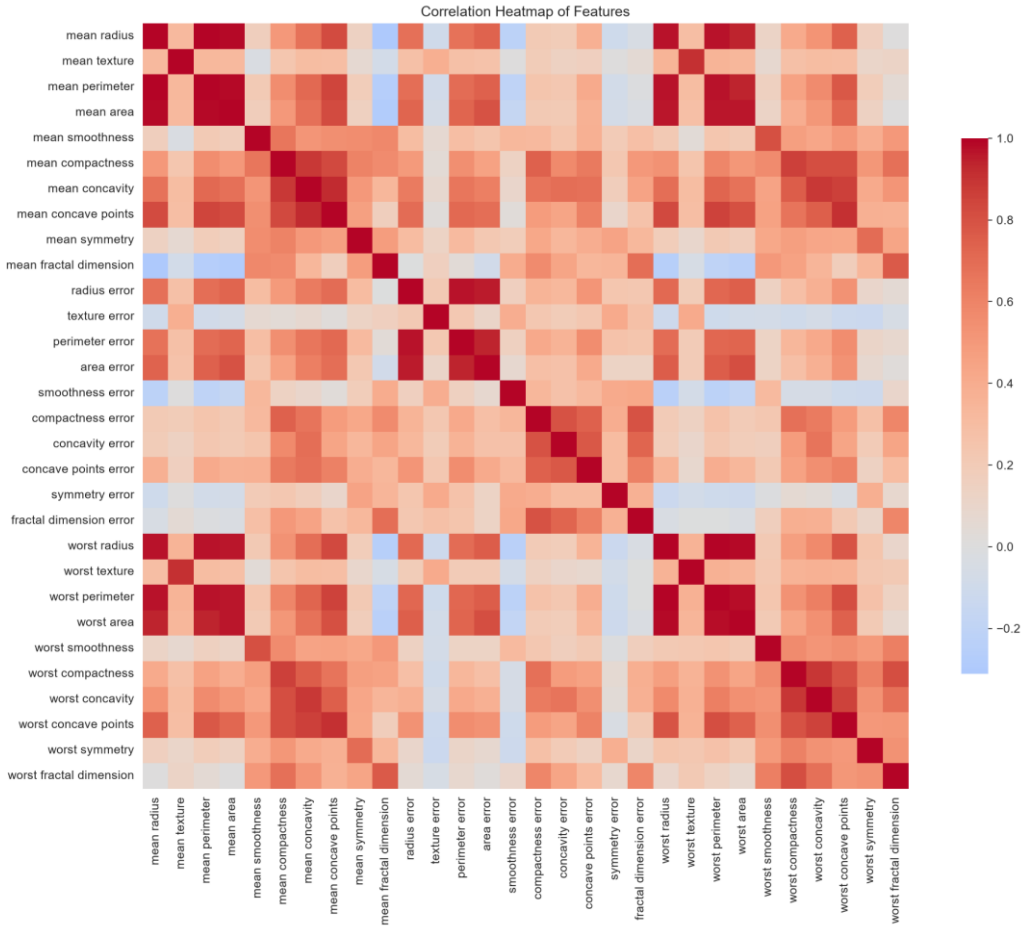


Figure 3: Correlation heatmap of the 30 features

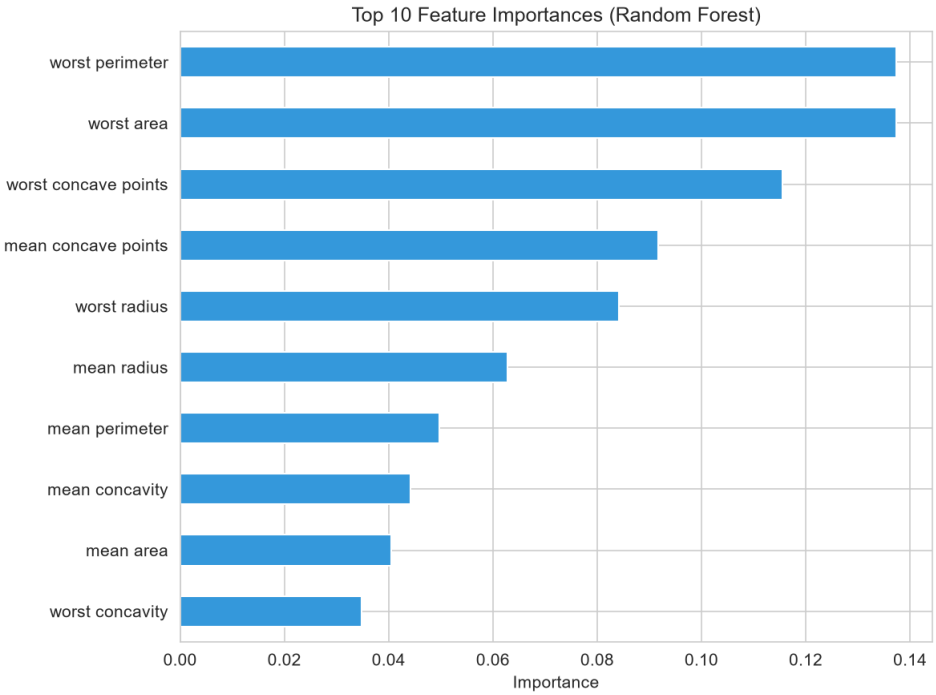


Figure 4: Top 10 feature importances (Random Forest)

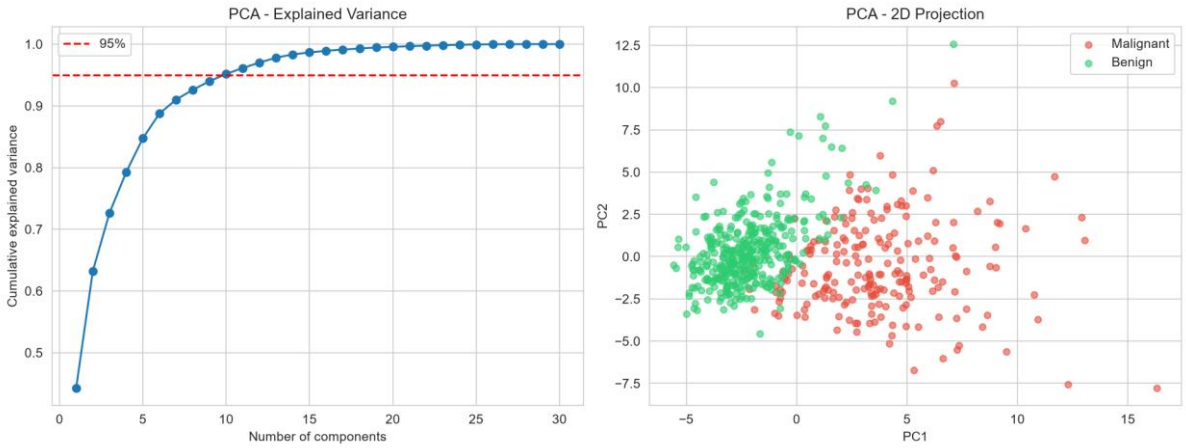


Figure 5: PCA, explained variance and 2D plot

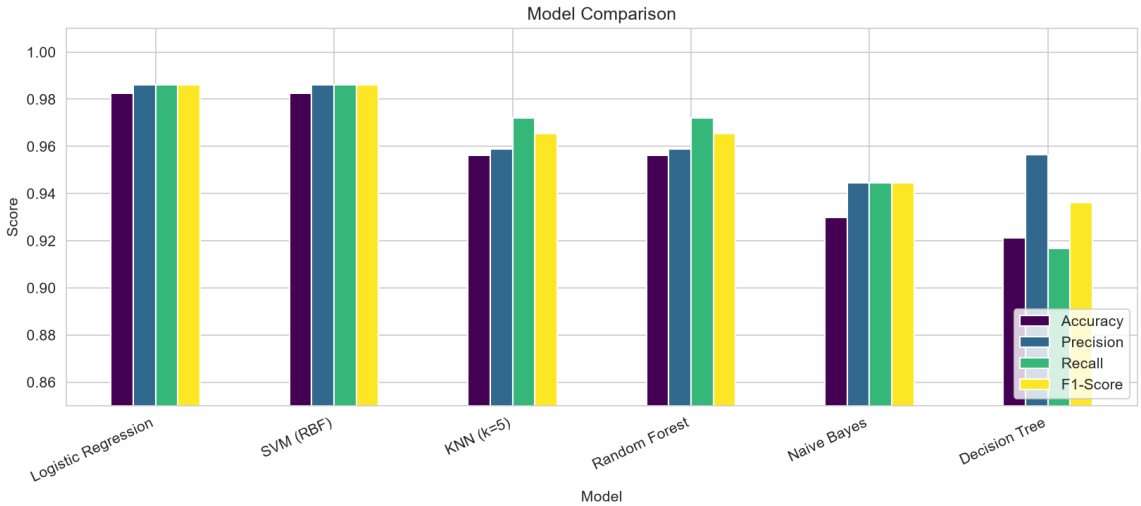


Figure 6: Comparison of the six models

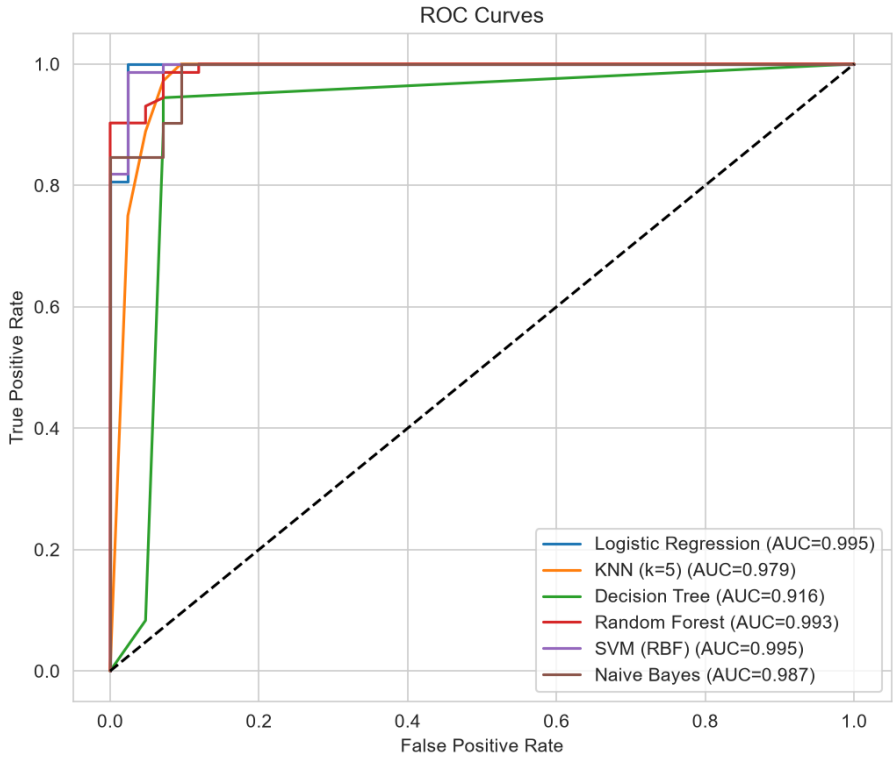


Figure 7: ROC curves of all models

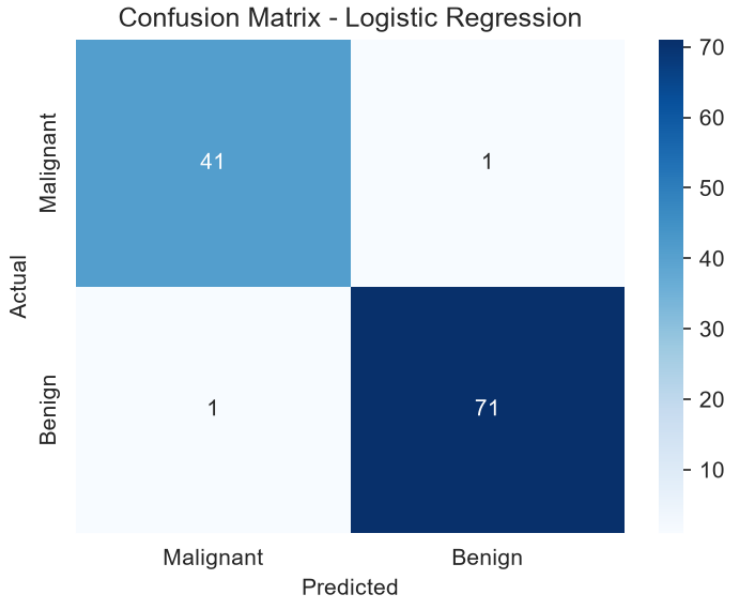


Figure 8: Confusion matrix of Logistic Regression

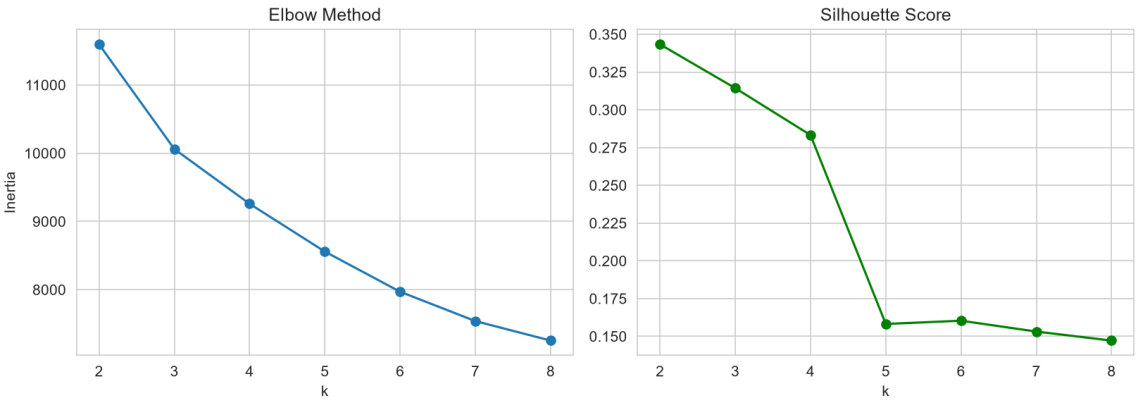


Figure 9: Elbow method and silhouette score for K-Means

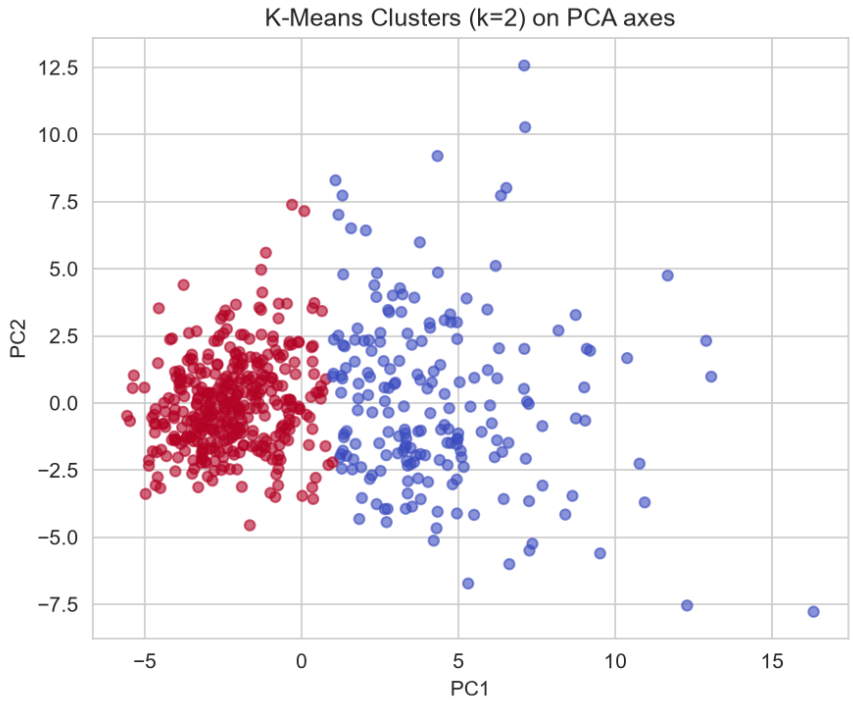


Figure 10: K-Means clusters (k = 2) on the PCA axes

8. Analysis

- The data has no missing values, so no cleaning was needed. 357 patients are benign and 212 are malignant (Figure 1).
- Malignant tumours have bigger radius, perimeter and area than benign ones (Figure 2). These three features are very closely related to each other (Figure 3).
- The most useful features were worst perimeter, worst area and worst concave points. Both methods gave the same top 10 features (Figure 4).
- With PCA, 2 components keep 63% of the information and 10 components keep 95% (Figure 5). In the 2D graph, malignant and benign patients are mostly separate.
- **Logistic Regression** and **SVM** gave the best result: **98.25% accuracy** (Table 1, Figures 6 and 7). Logistic Regression was also the best in cross-validation (97.8%).
- The Decision Tree was the lowest (92.1%). Random Forest, which uses many trees together, did better (95.6%).
- Logistic Regression made only 2 mistakes out of 114 test patients. In one of them, a cancer patient was predicted as benign (Table 2, Figure 8).
- K-Means, without using the diagnosis, still matched the real diagnosis for 90.5% of the patients (Table 3, Figure 10). The silhouette score was best for 2 clusters (Figure 9). But the supervised models are better than K-Means.
- Random Forest with only the top 10 features gave 93.9%, compared with 95.6% using all 30 features.
- Limits: the test data is small (114 patients), I did not tune the models, and the data is from one source. This project is only for learning and not for real medical use.

9. Conclusion

- Data mining can predict whether a tumour is cancer or not with very high accuracy on this dataset.
- The best models were Logistic Regression and SVM, with 98.25% accuracy.
- The size of the tumour (radius, perimeter, area) and the shape of its edge (concavity, concave points) are the most important features.
- In future, I can tune the models, try other models like XGBoost, and test on bigger data.

10. Sources and References

[1] Wolberg, W. H., Street, W. N., & Mangasarian, O. L. (1995). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5DW2B

[2] Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE Electronic Imaging, 1905, 861-870.

[3] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[4] Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.

[5] scikit-learn documentation: https://scikit-learn.org/stable/